Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/14/2022
PI Name:	Martin, Bryn Ph.D.		
Project Title:	Ophthalmic and Intracranial Structural Changes in Head-Down Tilt Bedrest: Potential Countermeasures and Comparison to SANS Findings in Astronauts (OPTICS study)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bmartin@neelatx.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	330-475-9747
Organization Name:	Neela Therapeutics		
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City:	Moscow	**State:**	ID
Zip Code:	83843	**Congressional District:**	1
Comments:	PI moved to Alcyone Therapeutics, Inc., in late 2020 from University of Idaho (Ed., December 2021) PI moved to Neela Therapeutics in late 2025 from Alcyone (Ed, November 2025)		
Project Type:	Ground	**Solicitation / Funding Source:**	2018-2019 HERO 80JSC018N0001-SANS: Spaceflight Associated Neuro-ocular Syndrome Countermeasures. Appendix C
Start Date:	12/08/2021	**End Date:**	12/07/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date of new grant 80NSSC22K0023 is 12/07/2024 per grant information documents (Ed., December 2021). NOTE: End date of previous grant number 80NSSC20K0920 changed to 10/12/2020 per JSC Grants Admin/NSSC information, due to PI move and new grant established (Ed., 12/10/21).		
Key Personnel Changes/Previous PI:	A postdoc has been hired for study data analysis and coordination, effective June 2022.		
COI Name (Institution):	Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Macias, Brandon Ph.D. (NASA Johnson Space Center) Marshall-Goebel, Karina Ph.D. (KBR/NASA Johnson Space Center) Williams, Michael M.D. (University of Washington, Seattle) Loerch, Linda Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC22K0023		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	NOTE: Continuation of "Ophthalmic and Intracranial Structural Changes in Head-Down Tilt Bedrest: Potential Countermeasures and Comparison to SANS Findings in Astronauts (OPTICS study)," grant 80NSSC20K0920, with the same Principal Investigator (PI) Dr. Bryn Martin. The proposed research plan builds on our group's ongoing research to quantify ophthalmic and intracranial changes in long duration spaceflight astronauts using novel non-invasive magnetic resonance imaging (MRI) protocols and post-processing techniques. We propose to use identical techniques currently implemented pre- and post-spaceflight to also quantify how long-duration bed rest and countermeasures applied during bed rest impact ophthalmic and intracranial structures. It is critically essential to use identical techniques that we have already applied in astronauts to determine if structural alterations seen in bed rest are similar to what our team has already quantified in astronauts with spaceflight associated neuro-ocular syndrome (SANS). Similarly, it is essential to use the identical techniques to quantify the capability of a countermeasure to mitigate structural alterations in bed rest.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will help better understand ophthalmic and cerebrospinal fluid physiology and anatomy that can potentially be used for diagnostic and therapeutic applications of terrestrial human diseases such as hydrocephalus and cerebrospinal fluid drug delivery.
Task Progress:	Spaceflight may potentially induce a bulk body fluid shift toward the head. Over long periods of time, such as in long-duration spaceflight (LDSF), the increase in fluid in the head may increase risk for conditions such as spaceflight-associated neuro-ocular syndrome (SANS) which can involve a flattening of the ocular orbit, and fluid shifts or volume changes of cerebrospinal fluid (CSF) and altered pressure acting on neural tissue. The OPTICS study is conducted on a group of healthy subjects undergoing a head-down tilt (HDT) procedure in an attempt to mimic fluid shifts observed in space and implement potential countermeasures to evaluate their potential efficacy in combatting the negative effects of LDSF on sensitive eye and brain tissues due to fluid shifts. Progress in the 2022 fiscal year involved subject recruitment and measure collection, as well as initial data analysis. Twenty-two subjects (11 men; 11 women, age range 24:52) were recruited and enrolled in study protocols for head-down tilt at the German Aerospace Center Institute for Aerospace Medicine (DLR) in Cologne, Germany. DLR is a state-of-the-art facility for conducting ground-based research in support of spaceflight. Study activities, including six-degree head-down tilt procedures to induce fluid shift toward the head, as an analogue for spaceflight, and potential countermeasures, including lower body negative pressure (LBNP, 25 mmHg ± 2 mmHG) commenced and magnetic resonance imaging of subjects was implemented for 30 days. Study data, including imaging data, underwent quality control and the study continues to be well underway. Initial data analyses have begun, including investigations into ophthalmic flattening, with the goal of determining if 1) head-down tilt procedures produce a similar flattening effect to spaceflight and 2) if applied countermeasures (LBNP) can counteract ophthalmic flattening induced in a head-down tilt environment. We found optic geometry changes, specifically flattening, in several subjects that participated in the study. It is yet unknown whether this alteration continues past the 30-day period of the study or whether countermeasures could impact this observed effect. Additional analyses for the study have begun, including the quantification of cerebral artery stroke volume (CAsv), peak-to-peak cerebral spinal fluid velocity in the cerebral aqueduct (CSFvp), and volume of the brain and CSF. When compared to a control group not exposed to a HDT condition there was an increase in intracranial volume (brain + CSF volume), with an increase in ventricular volume but not white matter volume. There was no observed effect of lower body negative pressure on these changes.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	G.C. Natividad, K. G. Warthen, S. H. Sater, B. R. Macias, S. Laurie, L. Pardon, B. A. Martin "Expanded measurements of MRI based ocular globe volume displacement in astronauts and precision assessment in controls" 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. Expanded measurements of MRI based ocular globe volume displacement in astronauts and precision assessment in controls. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	L. A. Kramer, K. H. Hasan, L. E. Bishop, X. Zhang, K. Marshall-Goebel, and B. A. Martin "Intracranial Effects of Intermittent Lower Body Negative Pressure with Head Down Tilt Bed Rest: Comparison to Upright Posture" 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. Intracranial Effects of Intermittent Lower Body Negative Pressure with Head Down Tilt Bed Rest: Comparison to Upright Posture. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	K. G. Warthen, G. Conley Natividad, S.H. Sater, L. A. Kramer, K. H. Hasan, M.A. Williams, K. Marshall-Goebel, B. R. Macias, S. Laurie, B. Caldwell, D. Gerlach, B.A. Martin "Optics Study: Optic Globe Volume Displacement due to Head Down Tilt Bedrest and Efficacy of Countermeasures" 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. Optics Study: Optic Globe Volume Displacement due to Head Down Tilt Bedrest and Efficacy of Countermeasures. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Sater SH, Conley Natividad G, Warthen K, Marshall-Goebel K, Laurie SS, Macias BR, Bershad EM, Sass AM, Seiner AJ, Kramer LA, Ethier CR, Fu AQ, Shrestha D, Rohr J, Stenger MB, Martin BA "MRI Quantification of Posterior Optic Globe Flattening in Head-Down Tilt Bed Rest Subjects and Long-Duration Spaceflight Astronauts" ISMRM Neurofluids Symposium (Rome, Italy, September 21-24, 2022). MRI Quantification of Posterior Optic Globe Flattening in Head-Down Tilt Bed Rest Subjects and Long-Duration Spaceflight Astronauts.ISMRM Neurofluids Symposium (Rome, Italy, September 21-24, 2022). , Sep-2022

Abstracts for Journals and Proceedings	Sater SH, Sass LR, Conley Natividad G, Arters O, Khani M, Watts R, Dobson H, Singh D, Martin BA "High Temporal Resolution Quantitative Contrast-enhanced MRI for Spatial-temporal Quantification of Cerebrospinal Fluid System Wide Intrathecal Gadolinium Pharmacokinetics in Nonhuman Primates" ISMRM Neurofluids Symposium (Rome, Italy, September 21-24, 2022). High Temporal Resolution Quantitative Contrast-enhanced MRI for Spatial-temporal Quantification of Cerebrospinal Fluid System Wide Intrathecal Gadolinium Pharmacokinetics in Nonhuman Primates," ISMRM Neurofluids Symposium (Rome, Italy, September 21-24, 2022). , Sep-2022
Abstracts for Journals and Proceedings	Martin BA, Sater SH, Marshall-Goebel K, Ploutz-Snyder RJ, Ethier CR, Stenger MB, Kramer LA, Macias BR "Automated MRI-Based Quantification of Posterior Ocular Globe Flattening and Recovery After Long-Duration Spaceflight" 2022 World Congress of Biomechanics (Taipei, Taiwan, July 10-14, 2022). Automated MRI-Based Quantification of Posterior Ocular Globe Flattening and Recovery After Long-Duration Spaceflight," 2022 World Congress of Biomechanics (Taipei, Taiwan, July 10-14, 2022). , Jul-2022
Abstracts for Journals and Proceedings	E.M. Bershad, K. Marshall-Goebel, S. S. Laurie, Scott Greenwald, L. Pardon, B. R. Macias, C.P. Venkatasubba Rao, B. A. Martin, S. H. Sater, H. Cohen, J.B. Clark, D. Kernagis, H. Sangi-Haghpeykar, G. E. Strangman, L.A. Kramer, K. M. Hasan, C. Stern, J. Rittweger, Stefan Moestl, and Jens Tank "Space-cent: Studying the Physiological and Anatomical Effects of Centrifugation and Head Down Tilt: 2021 Update" 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 8, 2022 "Space-cent: Studying the Physiological and Anatomical Effects of Centrifugation and Head Down Tilt: 2021 Update," 2022 NASA Human Research Program / IWS (Galveston, TX, February 8, 2022). , Feb-2022
Abstracts for Journals and Proceedings	Martin BA, Seiner A, Sater S, Natividad GC, Fu A, Macias BR, Laurie SS, Marshall-Goebel K, Kramer LA, Ploutz-Synder MB, Stenger MB "Ocular Health in Astronauts: MRI Quantification of Ocular Globe and Nerve Sheath Changes" 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 8, 2022 Ocular Health in Astronauts: MRI Quantification of Ocular Globe and Nerve Sheath Changes. 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 8, 2022 , Feb-2022
Articles in Peer-reviewed Journals	Stuart H Sater, Gabryel Conley Natividad, Akari J Seiner, Audrey Q Fu, Dev Shrestha, Eric M Bershad, Karina Marshall-Goebel, Steven S Laurie, Brandon R Macias, Bryn A Martin "MRI-based quantification of posterior ocular globe flattening during 60 days of strict 6° head-down tilt bed rest with and without daily centrifugation" J Appl Physiol (1985). 2022 Dec 1;133(6):1349-55. https://doi.org/10.1152/japplphysiol.00082.2022 ; PMID: 36326472; PMCID: PMC9744655 , Dec-2022